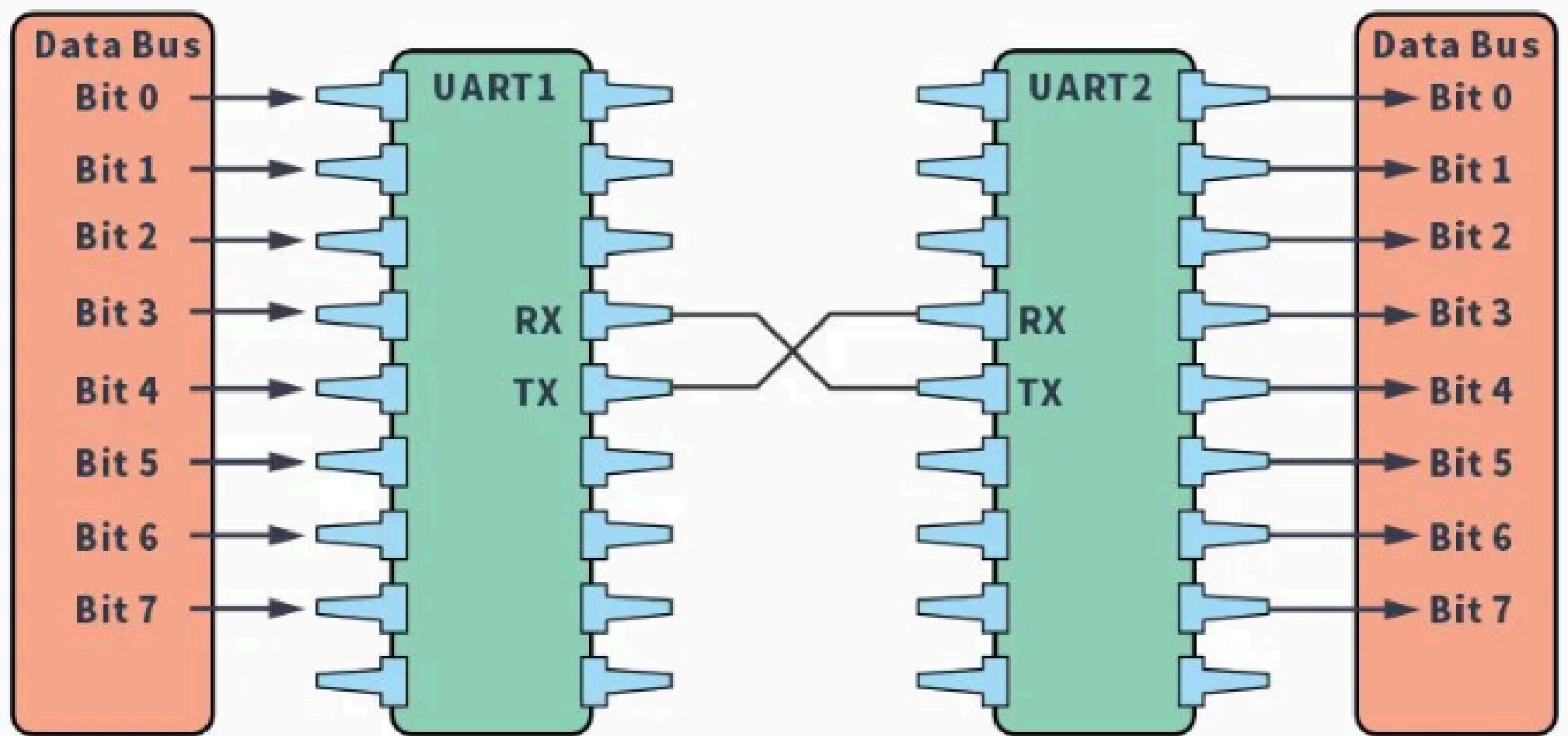


UART-TX

Communication Protocol



Prepared by / Nada Ahmed Mohamed Abdul-Kreem
Under Supervisor / Mohamed Elsayed

Table of Contents

- 1. Introduction**
- 2. Specification**
- 3. Architecture**
- 4. FlowChart**
- 5. State Diagram**
- 6. RTL Design**
- 7. Functional Simulation**
- 8. Synthesis & Mapping**
- 9. placement & routing**

1.Introduction

UART stands for **Universal Asynchronous Receiver–Transmitter**.

It is a hardware communication protocol that enables serial data exchange between two devices using just two data wires (**TX and RX**) and a shared ground

Unlike **SPI** and **I2C**, there is no shared clock signal.

Instead, both sender and receiver must independently know the communication speed (**Baud rate**) and agree on the data format before communication begins

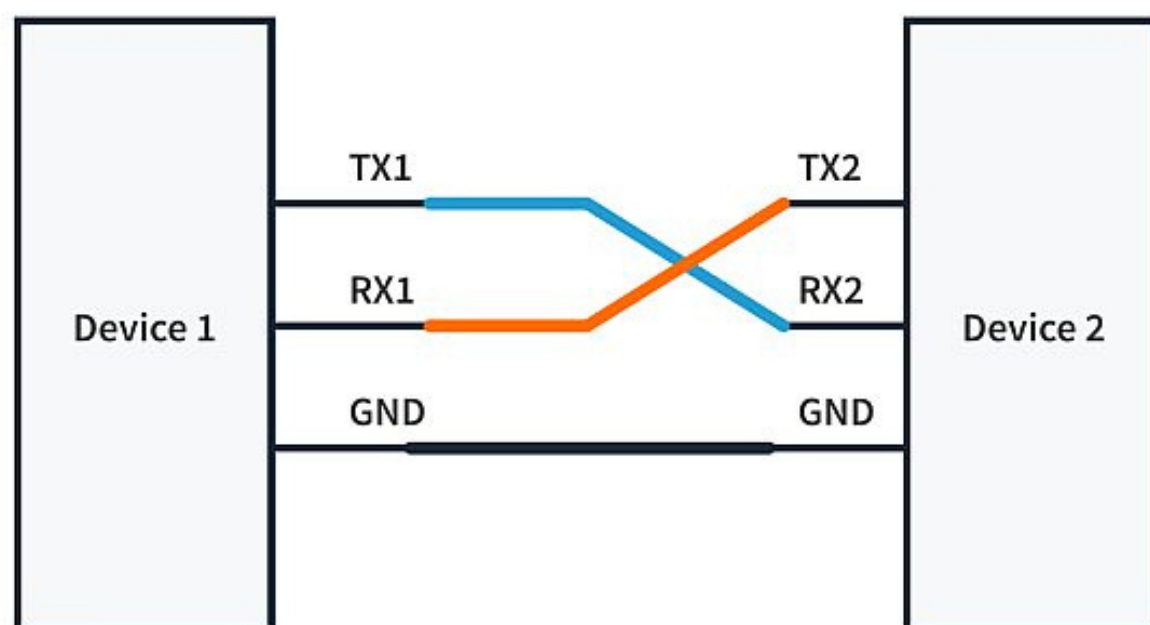
TX — Transmit

TX is the output line. The device drives this pin to send serial data. Bits are sent one at a time, with the least significant bit (LSB) first in standard UART.

RX — Receive

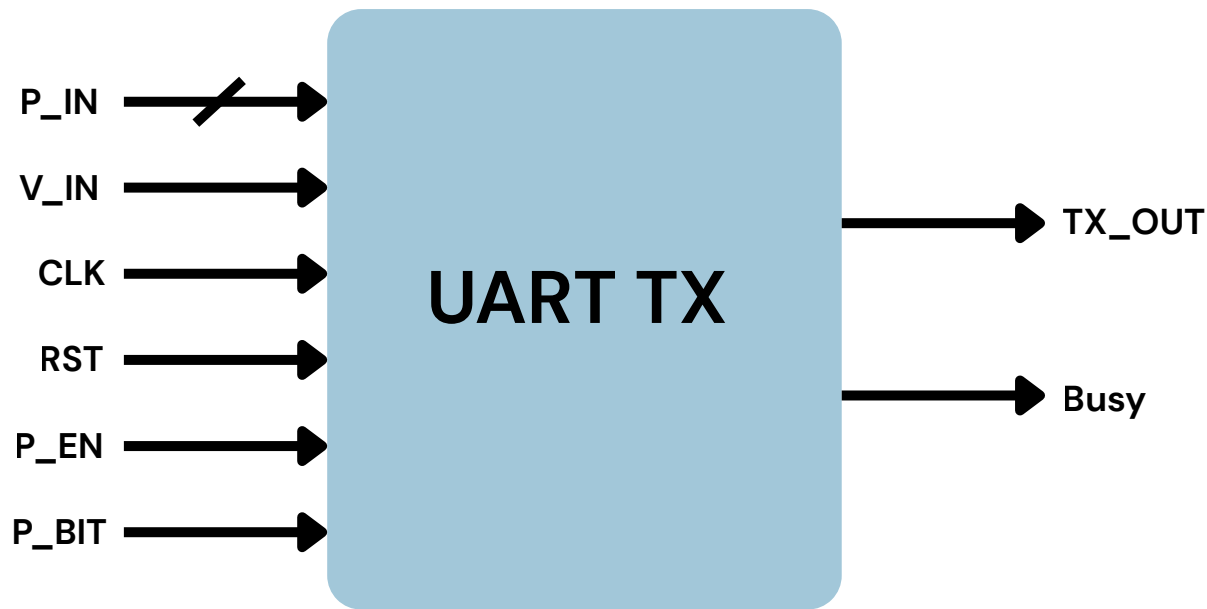
RX is the input line. The device monitors this pin to receive incoming serial data.

TX of device A connects to RX of device B, and RX of device A connects to TX of device B



2.Specification

Signal	Type	Description
P_IN	Input	Parallel input data to be transmitted
V_IN	Input	Valid signal; high for one clock cycle to request a new transmission
CLK	Input	System clock
RST	Input	Asynchronous active-low reset
P_EN	Input	Parity enable
P_BIT	Input	Parity type selection
TX_OUT	Input	Serial UART output
BUSY	Input	Indicates that UART TX is currently transmitting



- **Data Loading**

UART_TX receives new data from P_INPUT only when V_INPUT = 1.
V_IN is asserted for one clock cycle only.

- **Reset**

All internal registers are cleared using an asynchronous active-low reset.
When RST = 0, the UART TX returns to its initial state.

- **Busy Signal**

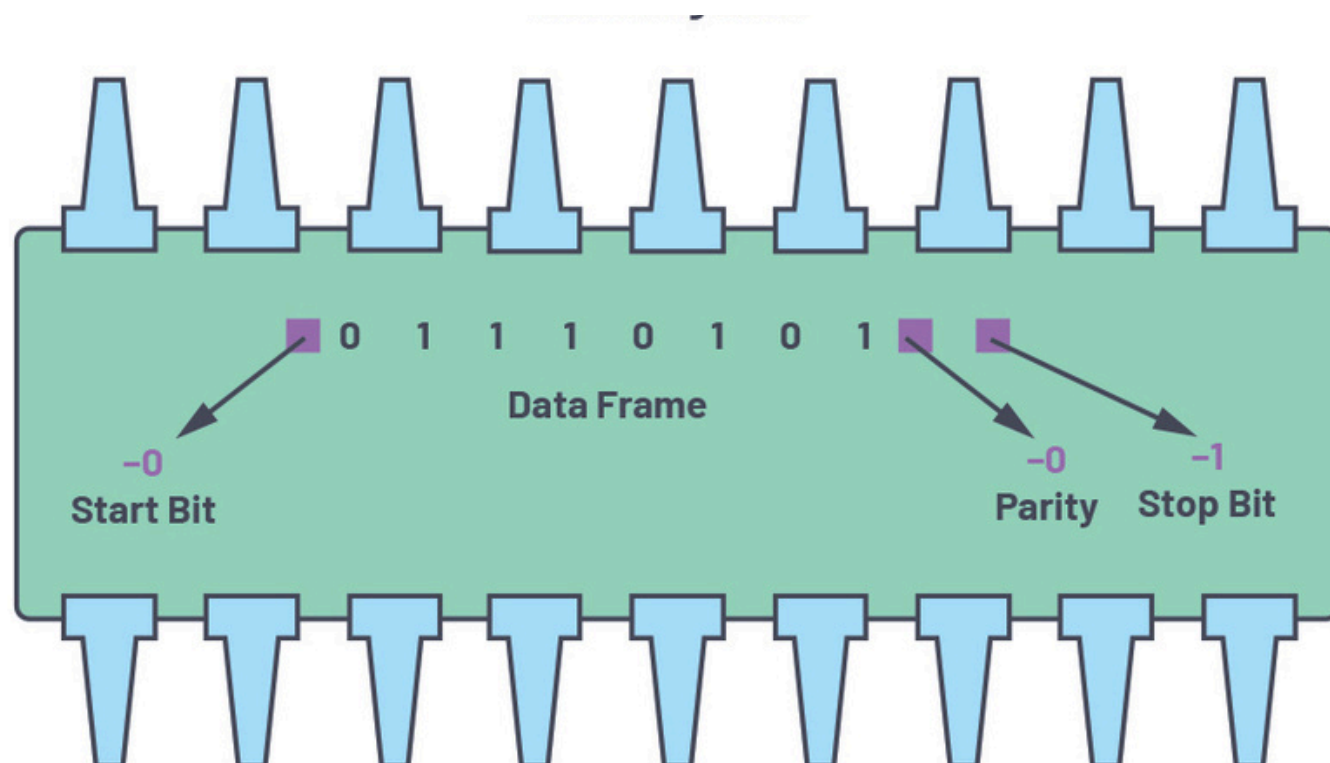
BUSY = 1 while the UART TX is transmitting the frame.
BUSY = 0 when transmission is complete.

- **Data Acceptance**

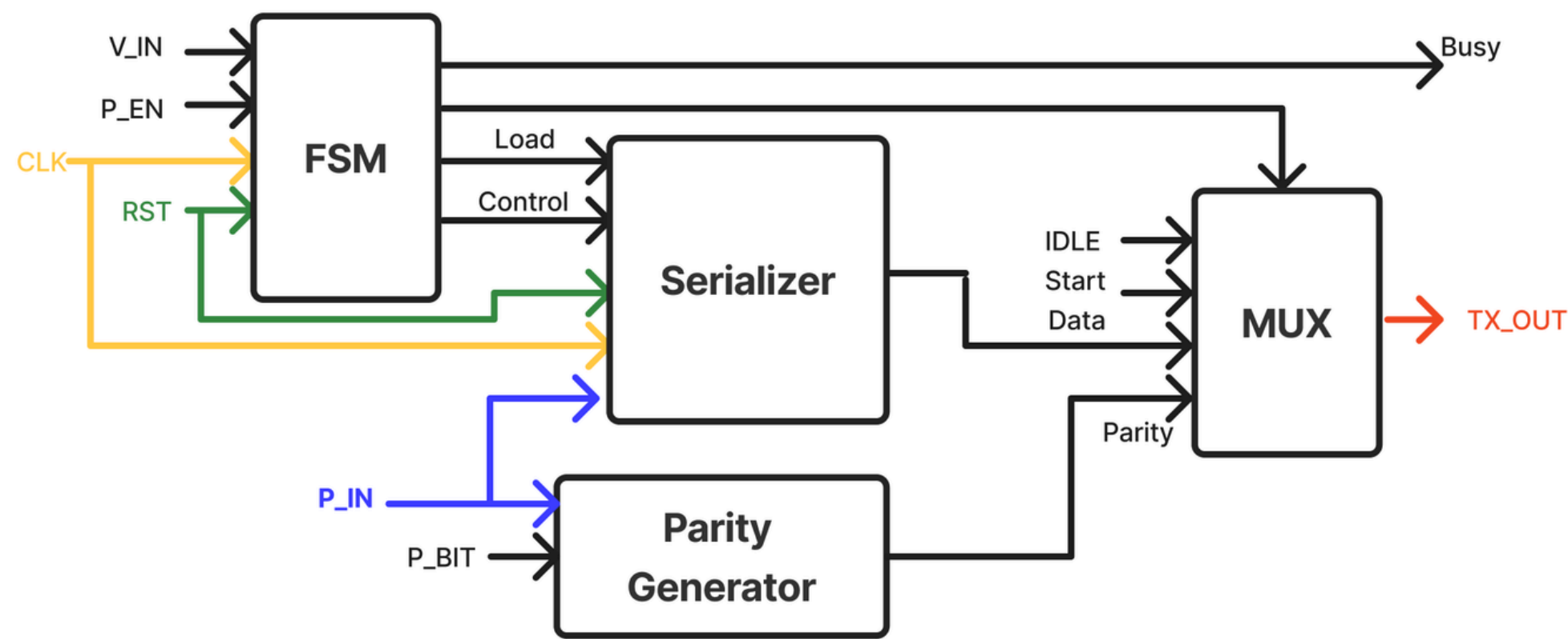
UART_TX can't accept new data while it is busy
New data can only be accepted when the transmitter is idle.

- **TX Output**

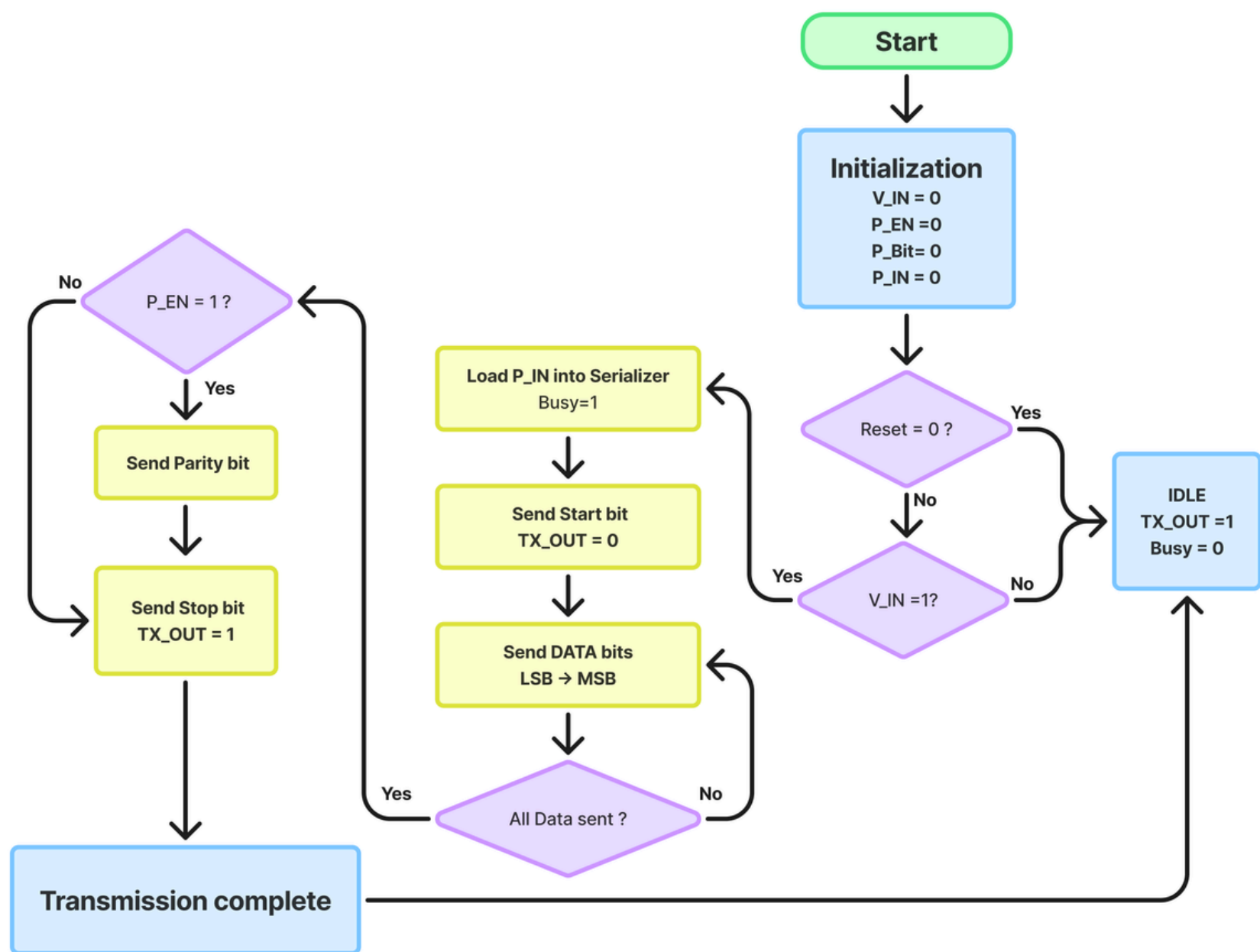
TX_OUT = 1 during the IDLE state.
The output then serially transmits the UART frame.



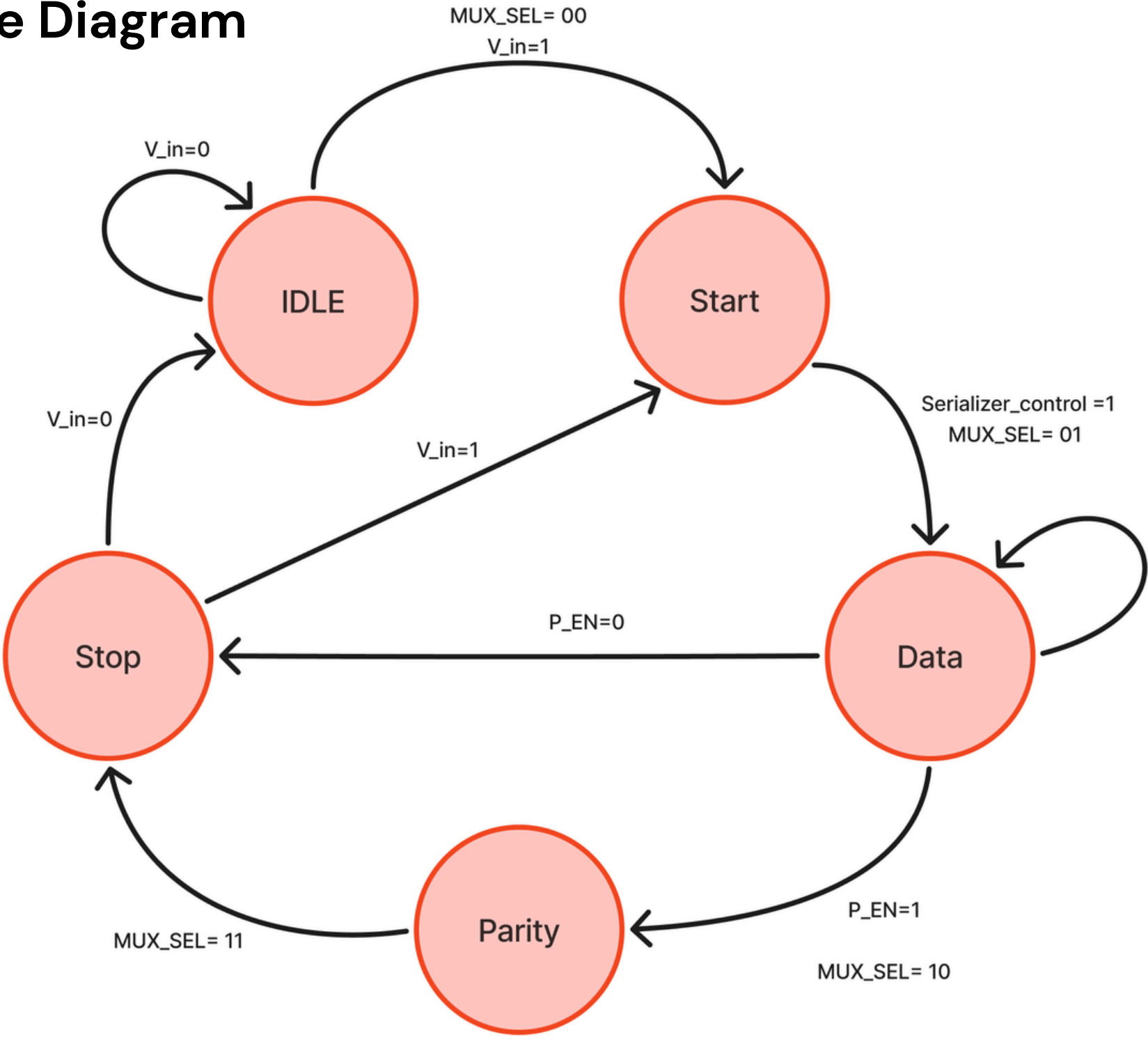
3.Architecture



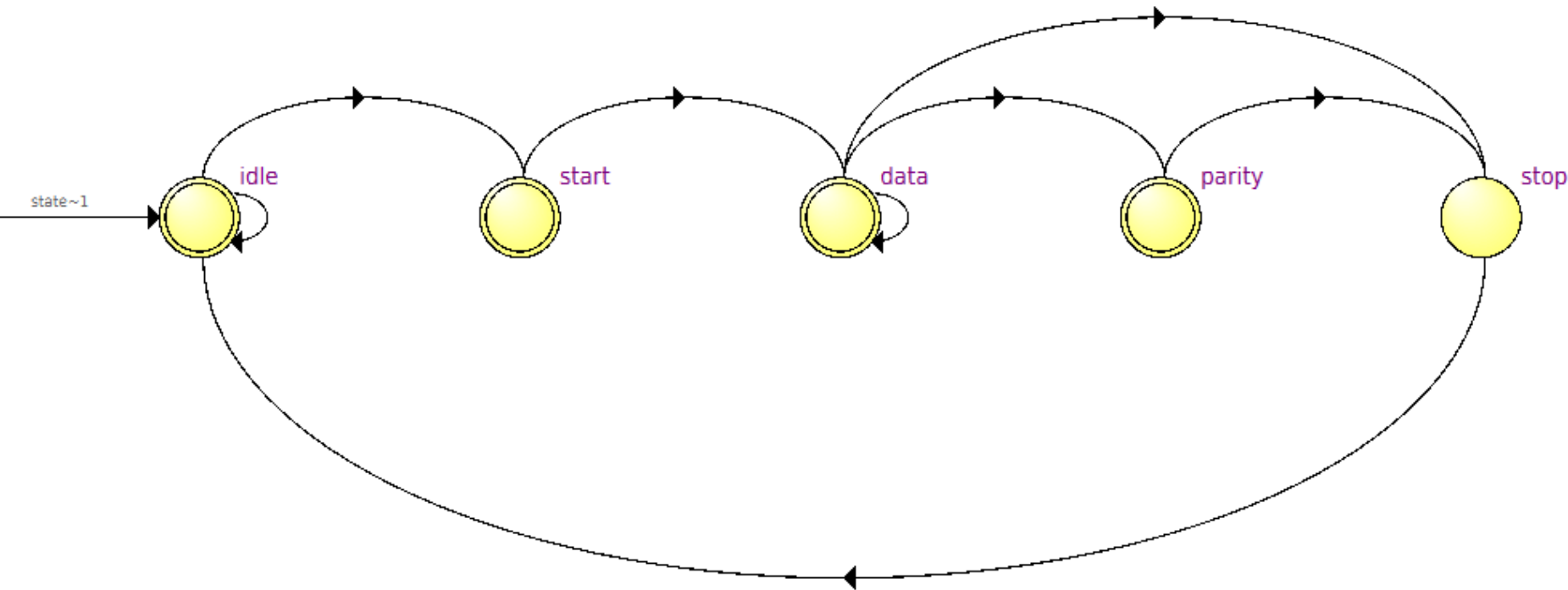
4.FlowChart



5.State Diagram



State Transition (Quartus)



6.RTL Design

- **FSM Controller**

The Controller is responsible for controlling the UART transmission sequence using a FSM. , It controls the transmission states, BUSY signal, serializer control, and MUX selection

IDLE → START → DATA → PARITY → STOP → IDLE

The PARITY state is skipped when P_EN = 0.

The controller prevents accepting new data while the transmitting

- **Serializer**

The Serializer converts the parallel input data into serial data.

Data is loaded when V_IN is high while the transmitter is IDLE.

The data bits are transmitted from LSB to MSB

- **Parity Bit Calculator**

P_EN enables or disables parity

P_BIT selects between even and odd parity

Then ,It generates the parity bit

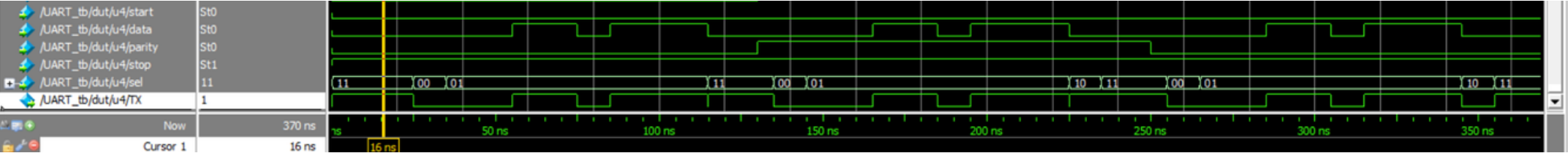
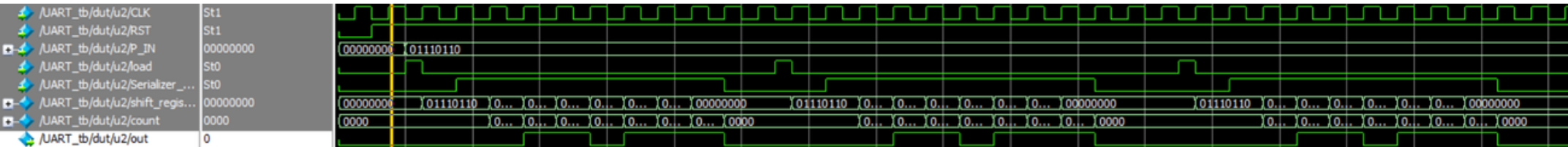
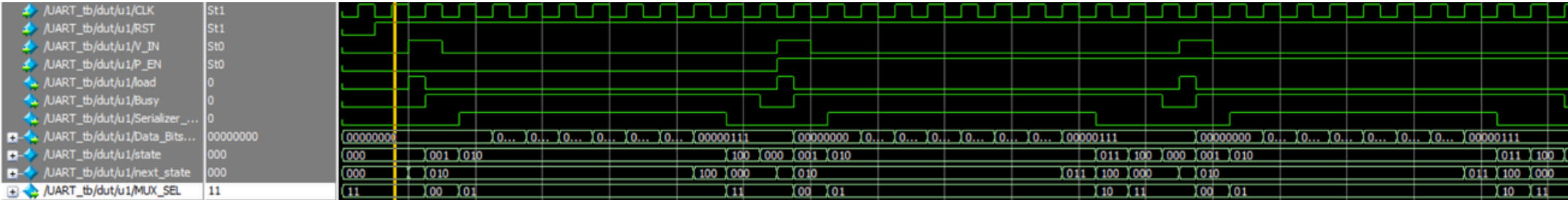
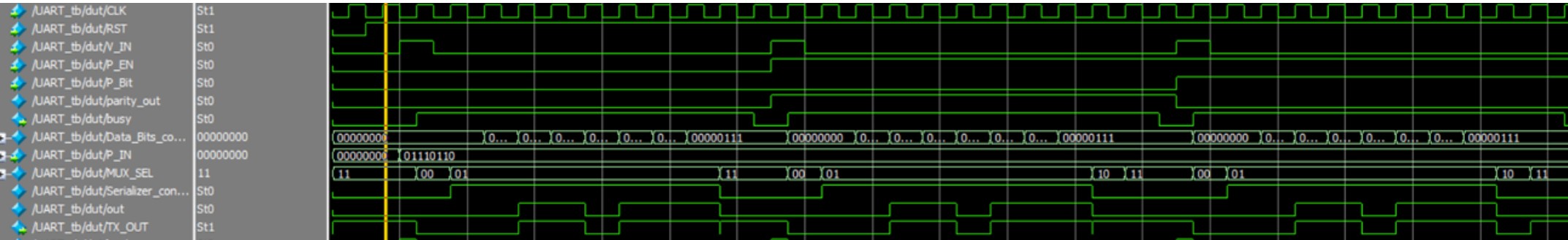
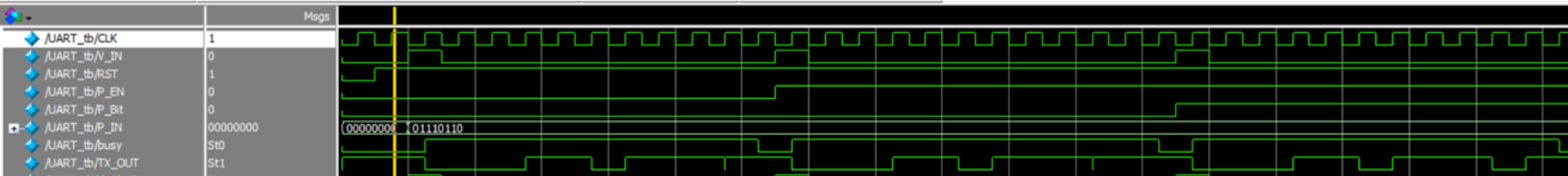
- **MUX**

selects the appropriate signal to drive TX_OUT according to the current FSM state.

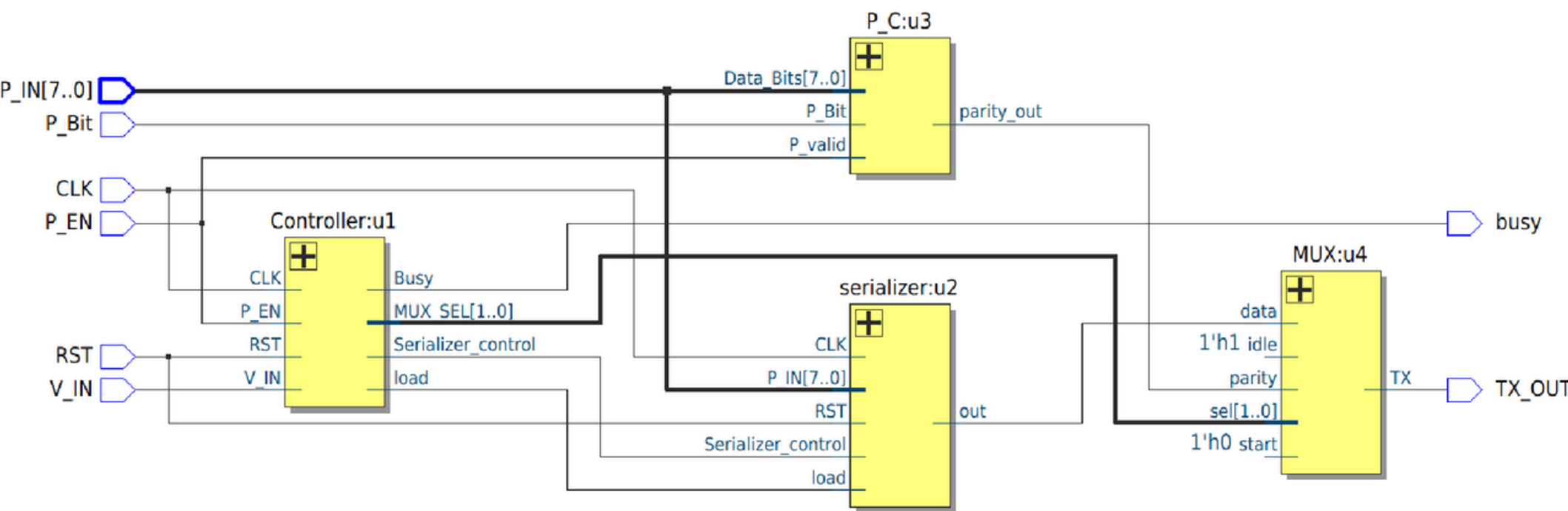
- **UART TX Top Module**

integrates all the internal modules and connects the controller, serializer, parity calculator, and MUX together

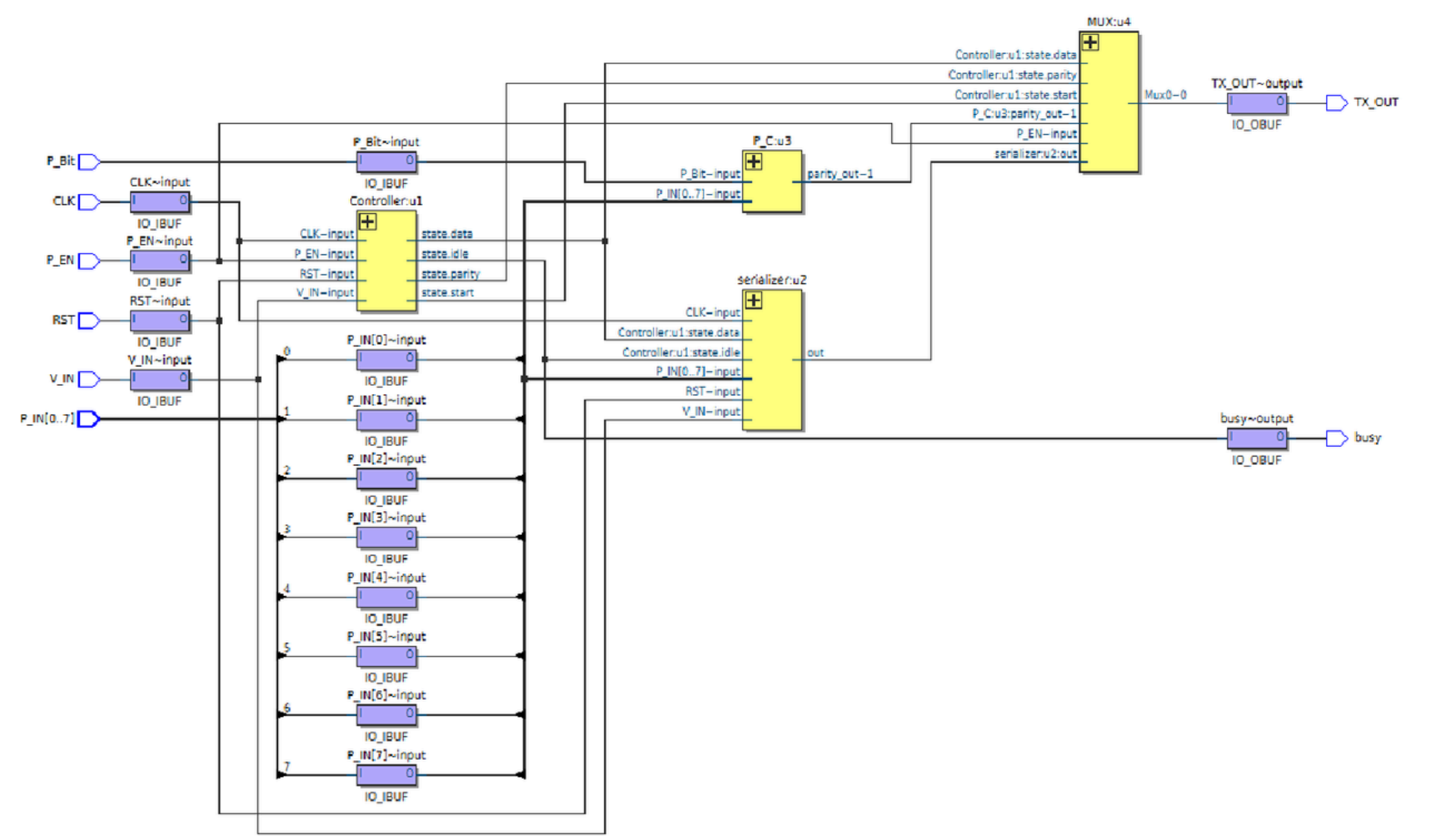
7.Functional Simulation



8.Synthesis



.Mapping



9.placement & routing

